

Digital Electronics

What digital electronic devices do you use?



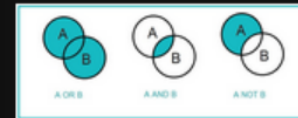
Boolean Algebra

Yes, No, Maybe...



- George Boole was a British mathematician born in 1815
- Devised a system of logic that aimed to condense complex thoughts into simple equations
- In Mathematics, algebra deals with finding the numeric value of a statement composed with variables that represent numbers
- Boolean algebra deals with determining the truth of a statement with true or false variables
- The Boolean logic (1854) laid the foundations for the digital revolution and paved the way for the computer age

Boolean Logic



- Three logic operators - AND, OR and NOT
- Operands defined by a set of two possible values - true/false, on/off, 1/0
- AND used to narrow the possible set of values capable of making a logic statement true
- OR broadens the possible set of values capable of making a logic statement true
- NOT excludes values from the possible set of values capable of making a logic statement true
- In a logic statement, NOT is evaluated first, AND is next and OR is last

{ Board Exercise }

Please stand up if the following logic statements are true:

1. girl AND brown hair AND left handed
2. girl OR brown hair OR left handed
3. girl AND NOT brown hair

{ Board Exercise }

Draw the truth tables for boolean operators

{ Exercise }

The Kitten Breeder



A real life problem

Let's imagine a cat breeder with very specific needs



There are all kinds of cats, but the options can be narrowed to:

- **M**ale, or **F**emale
- **W**hite, **B**lack, **T**an, or **O**ther
- and finally, **A**dults or **K**itties

Very specific needs

Everytime the breeder walks into a pet store he will ask:

I want a male kitty, either white or tan; or maybe a female kitty, any color but white; but I'll also settle for any cat you have as long as it's black.



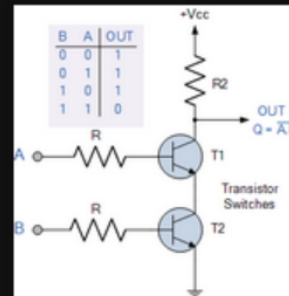
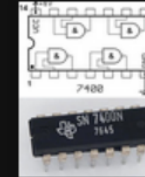
For a regular clerk this would be a nightmarish request.

But not for a boolean algebra expert!

Can you write down the expression representing this customer's wishes?

Transistor Switches

With two transistors a simple NAND gate can be implemented

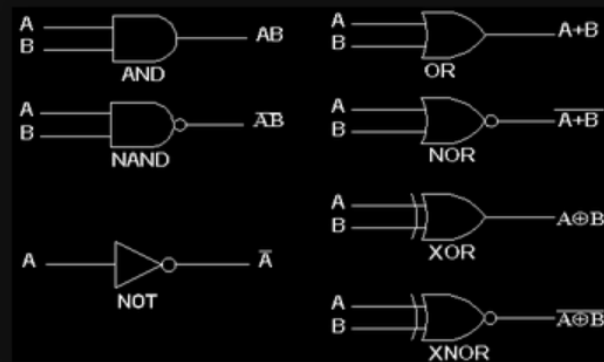


The NAND gate is important because any boolean function can be implemented by using a combination of NAND gates

One single IC (integrated circuit) can contain millions of transistors/logic gates

Logic Gates

Device implementing a Boolean logical operation on one or more binary inputs and producing a single binary output



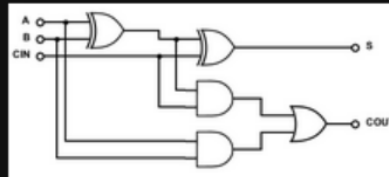
Relays (electromagnets), valves (vacuum tubes), or transistors can be used to implement logic gates

{ Board Exercise }

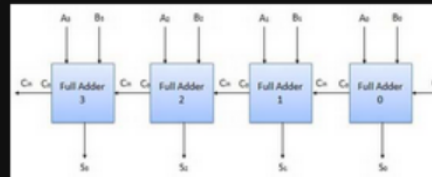
Build logic gates using only NAND gates

Digital Circuits

Logic Gates can be combined to produce complex circuits and logic



This is a full 1-bit adder. Four of these can be put together to form a full 4 bit adder



{ Exercise }

Back to the Kitten Store

Using any Logic Gates, draw the digital circuit that will solve the cat breeder's problem from the previous exercise

